

MECHANISMS OF "BUCCAL PUMPING" ("LUNG PACKING") AND ITS PULMONARY EFFECTS.

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INTRODUCTION

Increasing interest in breath hold diving and depth record breaking has brought attention to "lung packing" by "buccal pumping". Such manoeuvres have been used by spear fishing apnoea divers in the Mediterranean for many years as a mean to increase their depth capability. Hamilton (1) has described a technique used by US Navy divers to increase their VC. It was used more than 20 years ago by Croft, an ex-US Navy diver and record holder in apnoea diving at 73 m. Since buccal pumping is difficult to learn and not well described in the literature, only few subjects have been studied. Vann (2) measured the pressure-volume characteristics of the lungs of one subject after lung packing.

The aim of the present study was to further describe how buccal pumping is performed and to assess potential risks from lung packing. The risks involved are thought to be similar to accident mechanisms involved in pulmonary barotrauma in diving and free escape training, overinflation during mechanical ventilation, and positive pressure breathing in aviation.

METHODS

Three breath-hold divers volunteered for the study, which was approved by the local ethics committees. The divers were experienced in buccal pumping, and had participated in previous breath hold experiments.

Measuring cycle: Each subject made seven normal VC manoeuvres and seven VC followed by lung packing (VCL_p). To normalize lung history and circulation, the subjects stood up and made a deep knee bend after each measurement after which they rested sitting for five minutes before the next measurement. The number of buccal pump strokes or "chips" was individualized and adjusted so that each subject would be able to finish the slow staged exhalation without syncope. (The VCL_p volumes recorded are not to the maximal achievable.)

Spirometry: The relaxed airway pressures at different lung volumes above FRC were studied using a spirometer (Vicat 5, Mijnhardt, The Netherlands) fitted with an electric valve and pressure transducer (8510B-5, Endevco Corporation, CA, USA). The signals were fed to a PC using a custom made sampling program, which closed the mouthpiece valve after each 0.5 L exhalation, to allow measurement of the relaxed airway pressure. After investigator approval of the relaxed pressure recording, the valve was opened to allow further passive exhalation of 0.5 L. This procedure was repeated until FRC was reached and then the subject was asked to completely empty his lungs into the spirometer.

Radiology: To allow a good understanding of how buccal pumping is executed cineradiography of the pharynx and larynx in frontal and lateral view was performed using a hypopharynx software on a Siemens Polystar digital x-ray at 8 or 24 frames/s after the subjects had swallowed a mouth full of barium contrast (Mixobar High Density 2.4 mg/ml, Astra, Sweden).

Plain frontal and lateral chest x-rays on film at TLC and TLC_{Lp} were taken after all other procedures to elucidate how the additional gas volume after lung packing affected

chest expansion and also to detect possible mediastinal air after the series of 15-20 lung over-inflations in one day.

Cardiovascular data from the same series of experiments are presented in a companion paper (3).

RESULTS

On the x-ray VCR it can be seen that the diver starts with a mouth full of air and the tongue and bottom of the mouth in low position. With the lips and soft palate closed, the tip of the tongue, followed by the back of the tongue, are pressed upwards and finally the hypopharynx is squeezed to a small volume by lifting the larynx while the glottis stays open. Before the muscles are relaxed the glottis is closed and the oral cavity widened again to allow new air to enter the oral cavity through the lips. Note that it seems as if the epipharynx is closed and the epiglottis is open over the whole cycle. Trained persons can perform 1 - 2 such cycles or "chips" per second.

Table 1. Pulmonary parameters in three apnoea divers before and after lung packing (Lp).

Subj No	VC (L)		Incr.(%)	Volume (mL) pumpstroke	P airway _{relax} (kPa)		Insp.cap(L)	
	Norm	Lp			Norm	Lp	Norm	Lp
1	5.50	6.00	9	55	3.06	4.67	3.66	3.92
2	5.54	6.54	18	150	3.15	6.41	2.90	3.75
3	6.45	8.13	26	168	3.06	5.72	3.90	5.17

In our experiments eight to ten pumping movements or "chips" were used to fill the lungs with an additional 0.5 to 1.7 L. As much as 168 mL/ "chip" was recorded.

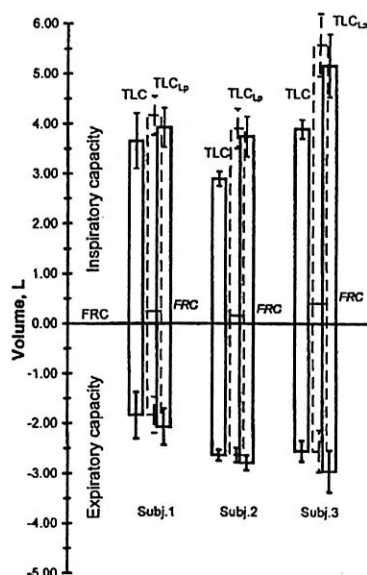


Fig 2. Inspiratory and expiratory capacities related to FRC (0 airway pressure), solid lines, and to unchanged RV, dotted lines.

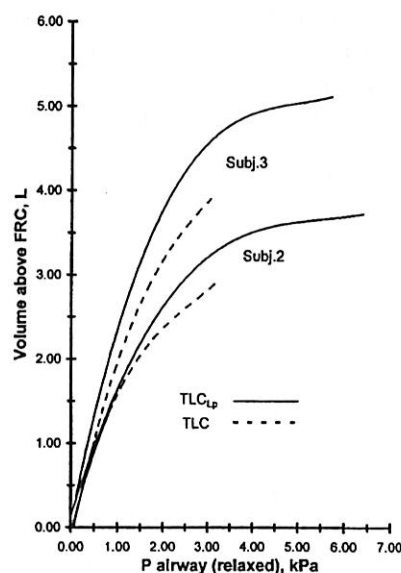


Fig 3. Passive relaxation curves starting from TLC and TLC_{Lp} in two subjects.

If zero airway pressure is defined as functional residual capacity (FRC) the inspiratory and expiratory capacities are distributed as the bars in Fig 2. If RV is assumed to be unchanged by previous lung packing the FRC (0 airway pressure) is increased.

In Fig 3 the lung volumes versus passive relaxation airway pressures are plotted for two subjects during stepwise passive exhalation from TLC and TLC_{LP}. One subject could not relax his glottis enough to give true airway pressures at the mouth.

No mediastinal gas could be found on the chest x-rays of any of the subjects after completion of the experiments. In all three subjects the additional air after buccal pumping (TLC_{LP}) gave less than one cm downward displacement of the diaphragm compared to the diaphragm position at TLC.

DISCUSSION

Theoretical maximum diving depth.

Published data on added volumes after lung packing amount to 22 - 39 % of VC (1,2, and 4), which corresponds well to volumes presented here. The ratio TLC to RV is one of the factors affecting the maximum diving depth. In a subject with a TLC of 5 L and an RV of 1 L a 20 % increase of the TLC by buccal pumping would increase the theoretical maximum depth by 25 %. However, blood shift to the thorax compensates for compressed gas and makes it difficult to make an accurate theoretical calculation of maximum diving depth. Anyway, in addition to the extra gas, a high intrathoracic pressure should keep the thoracic blood volume small at the start of the dive and thus allow more blood to enter during the dive at increased pressure, which should be an advantage if great depth is the goal.

Where does the extra air go?

Chest x-rays showed only minor deviations of the diaphragm comparing TLC and TLC_{LP}, and the method used did not allow more precise measurements of the rib cage movements.

If zero airway pressure is used as a reference to FRC volume it seems as if the ERV is larger after lung packing as indicated in Fig 2, solid line bars. This could be an effect of blood returning to the thorax after the high intrathoracic pressure during breath-hold at TLC_{LP}. An alternative interpretation is a reduced lung recoil after overinflation of the lungs and hence a larger FRC as indicated by the dotted line bars in Fig 2. Only measurements of lung compliance, FRC and thoracic blood/air-ratio could tell which of these interpretations is the right one.

Overinflation and lung rupture.

The risk for rupture of alveoli should be maximal when the trans-pulmonary pressure is high and the alveolar radius is great. A rupture might take place during forceful inspiration against a closed glottis during the mid part of inspiration with the most apical alveoli being at highest risk. When the normal TLC is reached the respiratory muscles are so short that not much of inspiratory force can be created. During the buccal pumping the glottis is closed at TLC and the respiratory muscles are relaxed, which increases the pleural pressure. The thorax is then at a volume greater than its relaxation volume and hence it supports the lung. The protective effect of the chest wall support (including its abdominal parts) on the expanding lung was shown by Schaefer et al (5) in dogs whose lungs could tolerate 24 kPa without damage when external support by thoracic and abdominal banding in comparison to 10 kPa when unsupported. In aviation 9 kPa positive pressure breathing is used to increase G-tolerance and to improve arterial oxygenation at extremely low ambient pressures. This is currently used with external counter pressure over both the thorax and the abdomen, but the risk for overinflation and rupture of the alveoli should not be overlooked (6). Not much

is known about the compliance of the thorax and the lungs at volumes above TLC, but it is speculated that the pressure increase per volume of the lung is steeper than for the unsupported thorax, which means that a critical wall tension of the alveoli will be reached sooner or later, because the thorax cannot support the lung with a sufficient external pressure. It seems most likely that the part of the chest wall which gives the least counter pressure is the diaphragm/abdominal wall. At which lung volume and transpulmonary pressure a rupture may happen can only be speculated on. Burst pressures between 8 and 10 kPa have been reported for lungs (5, 7), which means that divers using pulmonary packing may not be far from a severe damage of their lungs.

At the high absolute airway pressures (106 kPa) reached after lung packing at 100 kPa ambient pressure, the gas of an 8 L lung should be compressed about 0.5 L, which should be compensated for when trying to estimate the burst volume for the lung. This estimated compression is less than that found by Chaduteau et al (4) using full body plethysmography in connection to lung packing.

A 25 % overinflation corresponds to an ascent from 2.5 m with full lungs. According to literature an ascent from less than one meter with full lungs has been claimed as a definitive risk of pulmonary barotrauma and arterial embolism, and air embolization of cerebral arteries have even been observed in breath hold divers after a short dive to only about one m depth (8).

CONCLUSION

Lung packing while staying non-immersed at surface pressure creates an overinflation of the lungs, high intrathoracic pressures, and a potential risk of lung rupture. These effects are counteracted during the dive because of increased ambient pressure and compression of the lung volume. Nevertheless, lung overdistention after buccal pumping appears to incorporate a substantial risk for lung rupture.

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